

**ARCHITECT: THE ARCHITECTURE-BASED
TECHNOLOGY EVALUATION AND CAPABILITY
TRADEOFF METHOD**

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Presented to
The Academic Faculty

by

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In Partial Fulfillment
of the Requirements for the Degree
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To my son,

Alexander Daniel Cooksey

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LIST OF ACRONYMS

ABM Agent-Based Model

ARC-VM Architecture Real Options Complexity-Based Valuation Methodology

ARCHITECT Architecture-based Technology Evaluation and Capability Tradeoff
Method

ARCNET Architecture Resource-based Collaborative Network Evaluation Tool

C2 Command and Control

C4ISR Command, Control, Communications, Computers, Intelligence, Surveillance
and Reconnaissance

CADM Core Architecture Data Model

CBA Capabilities-Based Assessment

CCA Clinger Cohen Act

CDM Conceptual Data Model

CER Concept Exploration and Refinement

CONOPS Concept of Operations

CONUS Continental United States

CPM Critical Path Method

CV Capability Viewpoint

CVN Aircraft Carrier, Nuclear Variant

DAPS Defense Acquisition Program Support

DES Discrete Event Simulation

DIV Data and Information Viewpoint

DM2 DoDAF Meta Model

DoD Department of Defense

DoDAF Department of Defense Architecture Framework

DoE Design of Experiments

DOTMLPF Doctrine, Organization, Training, Materiel, Leadership and education,
Personnel and Facilities

DSM Design Structure Matrix

FDC Functional Domain Complexity

FOB Forward Operating Base

FOI Swedish National Defence Research Agency

FPC Functional Processing Complexity

G/Q/M Goal/Question/Metric

GAO General Accounting Office

HOQ House of Quality

ICAMS Integrated C4ISR Analysis and Management System

ICD Initial Capabilities Document

IOL Interoperability Level

IRMA Interactive Reconfigurable Matrix of Alternatives

IV Independent Variable

JCIDS Joint Capability Integration and Development System

JFCOM Joint Forces Command

KPP Key Performance Parameter

LDM Logical Data Model

LISI Levels of Information System Interoperability

M&S Modeling and Simulation

MC Markov Chain

MCS Monte Carlo Simulation

MDA Missile Defense Agency

MoA Matrix of Alternatives

MoE Measure of Effectiveness

MoFEs Measures of Force Effectiveness

MoP Measure of Performance

NVAC National Visualization and Analytics Center

O&S Operations and Support

OCONUS Outside of the Continental United States

OV Operational Viewpoint

PAID Procedures, Application, Infrastructure, and Data

PERT Program Evaluation and Review Technique

PES Physical Exchange Scheme

PSC Preferred System Concept

PSM Practical Systems/Software Measurement

PV Project Viewpoint

QFD Quality Function Deployment

RGS Requirements Generation System

ROSETTA Relational-Oriented Systems Engineering for Technology Tradeoff Analysis

RPC Resource Processing Complexity

RSC Resource State Complexity

RSE Response Surface Equation

RSM Response Surface Methodology

SD System Dynamics

SE Systems Engineering

SME Subject Matter Expert

SOF Special Operations Forces

SoS System of Systems

SoSE System of Systems Engineering

SPN Stochastic Petri Net

StdV Standards Viewpoint

SV Systems Viewpoint

SvcV Services Viewpoint

TD Technology Development

TRL Technology Readiness Level

V&V Verification and Validation

PREVIEW

SUMMARY

The use of architectures for the design, development, and documentation of system-of-systems engineering has become a common practice in recent years. This practice became mandatory in the defense industry in 2004 when the Department of Defense Architecture Framework (DoDAF) Promulgation Memo mandated that all Department of Defense (DoD) architectures must be DoDAF compliant. Despite this mandate, there has been significant confusion and a lack of consistency in the creation and the use of the architecture products. Products are typically created as static documents used for communication and documentation purposes that are difficult to change and do not support engineering design activities and acquisition decision making. At the same time, acquisition guidance has been recently reformed to move from the bottom-up approach of the Requirements Generation System (RGS) to the top-down approach mandated by the Joint Capabilities Integration and Development System (JCIDS), which requires the use of DoDAF to support acquisition. Defense agencies have had difficulty adjusting to this new policy, and are struggling to determine how to meet new acquisition requirements.

This research has developed the Architecture-based Technology Evaluation and Capability Tradeoff (ARCHITECT) Methodology to respond to these challenges and address concerns raised about the defense acquisition process, particularly the time required to implement parts of the process, the need to evaluate solutions across capability and mission areas, and the need to use a rigorous, traceable, repeatable method that utilizes modeling and simulation to better substantiate early-phase acquisition decisions. The objective is to create a capability-based systems engineering methodology for the early phases of design and acquisition (specifically Pre-Milestone